

Muhammad Alfiyandy HARIANSYAH

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 Room 6111, 6/F (Lift 19), Academic Building, HKUST, Clear Water Bay, Hong Kong
 Nationality : Indonesia  Indonesian, English, Japanese

An engineer and a student with a track record of developing machine learning techniques for aerodynamic design optimization and industrial experience in structural analysis. Currently pursuing his research interest in the conceptual design of eVTOL aircraft.

EDUCATION

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| 2024 - present | Ph.D. in Mechanical Engineering, HKUST, CGA : 4.30/4.00
Courses : Foundation of Solid Mechanics, Finite Element Methods, Data-Driven Modeling and Control of Dynamic Systems, Advanced Numerical Methods I.
Advisor : Prof. Rhea Liem |
| 2021 - 2023 | M.Eng. in Aerospace Engineering, Tohoku University, CGPA : 3.92/4.00 (Scholaro Link)
Courses : Mathematical Modeling and Computation, System Control Engineering, Numerical Analysis, Fluid Design Informatics, High Performance Computing, Fluid Dynamics, Structural Mechanics, etc.
Thesis : <i>Deep Learning Techniques for Aerodynamic Wing Shape Optimization</i>
Advisors : Prof. Shigeru Obayashi and Prof. Koji Shimoyama |
| 2017 - 2021 | B.Eng. in Mechanical and Aerospace Engineering, Tohoku University, CGPA : 3.96/4.00 (Scholaro Link)
Courses : Aircraft Design, Fluid Mechanics I and II, Theory of Elasticity, Computational Fluid Dynamics, etc.
Thesis : <i>An Artificial Neural Network-Assisted Genetic Algorithm with Application to Multi-Objective Transonic Airfoil Shape Optimization</i> . Advisor : Prof. Koji Shimoyama. |

EXPERIENCE

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|----------------------------------|--|
| February 2024 - present | PhD Researcher in Aircraft Design, HKUST, Hong Kong <ul style="list-style-type: none">➢ Achieved PhD candidacy after successfully passing the PhD Qualifying Examination (PQE), defending a research proposal on future-aware eVTOL design optimization and time-explicit LCA.➢ Developed a Hybrid MDF-IDF Multidisciplinary Design Optimization (MDO) framework in OpenMDAO to size and optimize Lift-Plus-Cruise and Multirotor eVTOLs, integrating aerodynamics, rotor performance, trim calculation, and weight estimation.➢ Formulated a multi-fidelity distributed iterative optimization architecture to coordinate system-level sizing with component-level detailed designs (e.g., VLM-based wing shape, BEMT-based rotor blade shape) via fixed-point physical parameter updates.➢ Pioneered a future-aware design methodology incorporating battery technology projections to determine cross-scenario robust designs and optimal discrete upgrade schedules using genetic algorithms.➢ Authored multiple research manuscripts detailing eVTOL MDO, technology forecasting, and system-component coordination, currently in preparation for submission to journals.➢ Maintained a perfect GPA (4.30/4.00) by achieving perfect scores (A+) in three advanced computational course projects : structural sizing of a PA-38 wing using one-way/two-way FSI (MECH 5930), RL-based adaptive PID pitch control using PPO (MECH 6910T), and a Roe-MUSCL finite-volume solver for 1D Euler equations (MATH 5311). <div style="display: flex; gap: 5px;">OpenMDAOOpenAeroStructANSYSPythonMachine LearningeVTOL Conceptual Aircraft Design Optimization</div> |
| Feb - May 2025
Feb - May 2026 | Graduate Teaching Assistant MECH 3620 - Aircraft Design, HKUST, Hong Kong <ul style="list-style-type: none">➢ Facilitated interactive tutorial sessions using Julia (Pluto.jl), instructing students on computational methods and programming for aircraft conceptual design.➢ Taught comprehensive tutorial topics including : Overall Aircraft Design through the AeroFuse framework, Introduction to Julia Programming, Constraint Analysis (Matching Chart), Preliminary Weight Estimation, Low-to-Mid Fidelity Fixed-Wing Aerodynamics (DATCOM and Vortex Lattice Method), Mission and Trim Analysis, Weight & Balance, Stability Analysis, and Velocity-Loads (V-n) Diagrams.➢ Co-authored the final group design project definitions (Request for Proposal) with Prof. Zhenyu Gao, formulating detailed mission profiles, performance constraints, and certification requirements for a narrow-body commercial jet (2025) and a high-efficiency regional jet (2026).➢ Evaluated and graded student design projects : assessing midterm and final presentations, as well as comprehensive technical reports based on technical quality, trade-off analyses, and design rationales.➢ Provided continuous technical support and project consultation to student groups. <div style="display: flex; gap: 5px;">AeroFuseOpenVSPXFLR5XFOILJuliaPluto.jlLaTeX</div> |

August 2023 -
January 2024

Structural Analysis Engineer (full-time) | PT Turkish Aerospace Indonesia, BANDUNG, Indonesia

- Contributed as an analysis engineer for two projects : 1. Weight Reduction for ATA 27 Components (Flight Control System) of the TAI Hürjet (Advanced Jet Trainer), 2. Preliminary Design Study of the Horizontal Stabilizer for the TAI T625 Gökbey (Transport and Light Utility Helicopter).
- Performed static (SOL 101) and normal mode (SOL 103) analyses to determine structural integrity (Reserve Factor) using finite element methods (MSC Patran/Nastran) backed with hand calculations.
- Responsible for analyzing ATA 27 Components : 1) Rudder (Installation, Mounting Bracket, Lever Arm), 2) LEFAS Assy. (PDU Instl., Control Module Fitting, Bevel Gear), 3) Flaperon Instl., 4) IMU Instl.
- Performed hand calculation analyses for : 1) bolted joint, 2) lug and pin, 3) flanged stud.
- Prepared FEM : Load and BCs, 3D Mesh TET10, SPC, MPC (RBE2 vs RBE3), CBUSH, CGAP, etc.
- Automated the application of MPC and CBUSH using scripting in Patran Command Language.
- Generated detailed documentation for modification in designs and their analysis reports.
- Performed aerodynamic analysis of the HorzStab using the Schrenk method, Vortex Lattice Method, and 3D Panel method to provide spanwise and chordwise load distribution.
- Calculated the shear force, bending moment, and torsion due to aerodynamic forces on the HorzStab.
- Wrote pyNastran code to extract forces/stresses automatically from HorzStab GFEM via OP2 files.
- Joined training about regulations : DOA (CS-21 Subpart J) and CS-29 for large helicopter.
- Presented in several weekly meetings with the team in the Turkish Aerospace Industries.

MSC Patran Nastran Patran Command Language Hypermesh PyNastran OpenVSP Python MS Office

August 2022 -
July 2023

Aerodynamics Engineer (part-time) | teTra aviation corp., Tokyo, Japan

- Performed a multi-objective design exploration study of a tandem wing configuration to maximize aerodynamic performances and meet stability requirements for a forward flight of our eVTOL (Mk-5).
- Spearheaded the development of an aircraft-level integration environment (aerodynamics, structures, propulsion, weights, controls) using the SUAVE framework (from ADL Stanford University).
- Spearheaded the development of a tool for airfoil exploration and selection (1500+ airfoils database).
- Performed multi-fidelity subsonic aerodynamic analyses of eVTOL fixed-wing multi-rotor systems.
- Managed simulation portfolio (3D geometry, mesh, CFD data, automation pipeline, etc.)
- Performed conceptual design (eVTOL iterative sizing, weight breakdown analysis, required battery power over an intended flight profile, reversed engineering process, what-if analyses, etc.)
- Assisted manned flight test preparation at Fukushima RTF (operation, flight log analysis, PID tuning).
- Wrote technical reports, drew conclusions, and made suggestions.

Python Linux OpenVSP Xflr5 XFOIL ANSYS Fluent SUAVE Flight Review (PX4) Notion

April 2020 -
September 2023

Student Researcher | Fluids Engineering with Data Science Laboratory, TOHOKU UNIVERSITY, Japan

- Worked on collaborative research projects under “The Program for Promoting Research on the Super-computer Fugaku” by MEXT and “Boeing Higher Education Program” by the Boeing Japan.
- Developed an in-house surrogate-based optimization framework that utilizes a dynamically retrained multilayer perceptron combined with a genetic optimizer.
- Applied the framework to aerodynamic design optimization of 2D and 3D transonic wings : PARSEC, B-spline airfoils, and NASA Common Research Model (CRM) wing.
- Developed a DCGAN-based generative method to produce synthetic wing designs and a CNN-based geometric filtering method to filter abnormal shapes and efficiently explore the design space.
- Collaborated with researchers at IFS to apply the framework to wing structural layout optimization.
- Automated the geometry production, meshing, CFD analysis, and optimization on an HPC system.
- Presented the research results at several domestic and international conferences.
- Presented seminars to other lab members on multi-objective optimization and CFD techniques.

Python C Linux HPC SU2 Pointwise MACH-Aero Tecplot PyTorch Pymoo

SKILLS

Programming Languages	Proficient (Python, C, Matlab/Simulink); Knowledgeable (C++, HTML5, CSS3)
Operating System and SCM	MacOS, Microsoft Windows, Linux Ubuntu; Git, GitHub, GitLab, Sublime Merge
Text Editing and Documentation	Sublime Text, VIM, Jupyter Notebook, Microsoft Word, Notion, LaTeX Overleaf
ML and Data Science	NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, TensorFlow, Spreadsheet
CFD and Meshing	ANSYS Fluent, ADflow, SU2, Pointwise, OpenAeroStruct
Finite Element Analysis	MSC Patran, Nastran, Hypermesh, PyNastran, ANSYS Mechanical
3D Geometry and Visualization	OpenVSP, PyGeo, SolidWorks, FreeCAD; Matplotlib, Plotly, Tecplot, Paraview
Aircraft Conceptual Design	OpenVSP, Flow5, Xflr5, XFOIL, XROTOR, SUAVE, AeroFuse
Multidisciplinary Design Optimization	OpenMDAO, ANSYS optiSLang, Pymoo
Flight Control and Analysis	Knowledgeable (Flight Review); Informed (PX4, jMAVSim, QGroundControl)
Language Proficiency	Professional (English : iBT 107/120); Conversational (Japanese N3); Native (Indonesian)

PROJECTS

MCEVS : MULTI-CONFIGURATIONAL EVTOL SIZING

JUNE 2024 - PRESENT

 [GitHub Repository](#)

Developing a sizing and optimization framework for eVTOL configurations. An ongoing PhD project advised by Prof. Rhea Liem.

WORK PACKAGE : WEIGHT REDUCTION FOR ATA27 COMPONENTS OF THE TAI HÜRJET

AUG 2023 - NOV 2024

 [PT Turkish Aerospace Indonesia's Portfolio](#)

Contributed as a structural analyst for the structures of the ATA 27 Components of an Advanced Jet Trainer.

DEEP LEARNING TECHNIQUES FOR AERODYNAMIC WING SHAPE OPTIMIZATION

2021 - 2022

 [Boeing Higher Education \(BHE\) Program organized by Prof. Shigeru Obayashi](#)

Conducted research funded by the BHE Program and reported the results to Mr. Will Shaffer, the President of Boeing Japan.

LEADING RESEARCH ON INNOVATIVE AIRCRAFT DESIGN TECHNOLOGIES TO REPLACE FLIGHT TEST

2020 - 2022

 [Fugaku Supercomputer Project](#)

Contributed research studies with Prof. Koji Shimoyama on AI applications in aircraft design (led by Prof. Soshi Kawai).

PUBLICATIONS

Articles

- Hariansyah, M. A., and Shimoyama, K., "An Artificial Neural Network-Assisted Genetic Algorithm With Application to Multi-Objective Transonic Airfoil Shape Optimization," *JAXA Special Publication : Proceedings of the 53rd Fluid Dynamics Conference/39th Aerospace Numerical Simulation Symposium*, 2022, pp. 115-124, JAXA-SP-21-008, ISSN 2433-2232.
- Hariansyah, M. A., and Shimoyama, K., "On the Use of a Multilayer Perceptron Based Surrogate Model in Evolutionary Optimization," *Proceedings of the Computational Mechanics Conference*, 2021, Vol. 2021.34, Online ISSN 2424-2799, DOI:10.1299/jsmecmd.2021.34.235

Presentations

- Hariansyah, M. A., and Liem, R. P., "A Physics-Driven Multifidelity Distributed Iterative Optimization Framework for eVTOL Aircraft Design," *14th ASMO UK/3rd ASMO-EUROPE/ISSMO Conference on Engineering Design Optimization*, Oral Presentation, July 2026, Imperial College London, London, United Kingdom.
- Hariansyah, M. A., and Shimoyama, K., "Deep Learning Techniques for High-Dimensional Surrogate-Based Aerodynamic Design," *33rd Congress of the International Council of the Aeronautical Sciences*, Oral, September 2022, Stockholm, Sweden.
- Hariansyah, M. A., and Shimoyama, K., "Aerodynamic Wing Shape Optimization via Deep Learning-Assisted Genetic Algorithm" *JSME Annual Meeting 2022*, Oral, September 2022, Toyama, Japan.
- Inaba, Y., Date, S., Hariansyah, M. A., Abe, Y., Shimoyama, K., Okabe, T., and Obayashi, S., "Optimization of Structural Layout for Composite Aircraft Wings," *the 18th International Conference on Flow Dynamics*, Online Poster Session, 2021.
- Hariansyah, M. A., and Shimoyama, K., "On the Use of a Multilayer Perceptron as an Aerodynamic Performance Approximator in Multi-Objective Transonic Airfoil Shape Optimization," *the 18th International Conference on Flow Dynamics*, 2021.

HONORS & AWARDS

January 2024	Hong Kong PhD Fellowship Scheme (prestigious program from the University Grants Committee of HK)
June 2022	IFS Graduate Student Overseas Presentation Award (a travel grant 350k JPY)
2021 - 2023	Mizuho International Foundation Scholarship Awardee (Top 15/ 60+ international applicants).
2021 - 2022	Boeing Higher Education (BHE) Program Student Research Project Awardee. (JPY 300k grant)
October 2021	Best Presentation Award at the 18th International Conference on Flow Dynamics (of 71 papers).
2017 - 2021	Japanese Government (MEXT) Scholarship Awardee (Top 10/100+ applicants globally).
May 2015	Gold Medal at Physics National Science Olympiad for Senior HS Students in Indonesia. (4th/99)
May 2013	Gold Medal (absolute) at Physics National Science Olympiad for Junior HS Students in Indonesia. (1st/99)
July 2012	Silver Medal at Physics National Science Olympiad for Junior HS Students in Indonesia. (8th/99)
December 2012	Finalist at the International Junior and Science Olympiad (IJSO) in Tehran, Iran.